ARKA AI

Intelligent Edge-AI Monitoring & Performance Optimization System for Solar Power Plants

1. Title of project/product

ARKA AI – Intelligent Edge-AI Monitoring & Performance Optimization System for Solar Power Plants

2. Problem Statement

Solar power plants operate under continuously changing irradiance, temperature and electrical conditions. Panel underperformance, abnormal voltage/current, inverter abnormalities and other performance deviations can remain unnoticed when monitoring is fragmented or delayed, reducing the effectiveness, reliability and usable output of renewable-energy generation.

3. Solution Proposed

ARKA AI adapts the core AquaMind architecture to solar power plants while preserving its basic sense–analyse–predict–respond model. Distributed sensor nodes based on ESP32-S3 collect **solar irradiance, panel temperature, voltage and current**, together with relevant environmental and inverter parameters. Voltage and current sensing is integrated at the electrical-output side of the monitored solar system; the measured values are sent by the ESP32-S3 through MQTT to a local gateway and Python/PySide6 HMI. The dashboard displays **live voltage (V), current (A), calculated power (W), energy generated (Wh/kWh), solar irradiance, temperature and historical trends**. Edge-AI pattern analysis compares observed behaviour with expected operating patterns, identifies anomalies and supports predictive maintenance and performance optimization. For an auxiliary sensor-node battery, a battery-voltage sensing channel can additionally report battery condition/life on the dashboard. Local storage enables operation and telemetry retention without continuous cloud dependence, with configurable alerts and response outputs.

4. Implementation Plan

Deploy sensor nodes across representative solar-panel/inverter zones; connect voltage/current and environmental sensors to ESP32-S3 nodes; transmit measurements using MQTT to a local gateway; display real-time electrical and environmental parameters on the ARKA AI dashboard; establish baseline operating patterns; validate anomaly detection against controlled and field conditions; and progressively integrate plant-level monitoring. **Target beneficiaries:** solar-plant operators, maintenance teams and renewable-energy asset owners. **Expected outcomes:** earlier identification of electrical and performance abnormalities, improved visibility of actual plant output, better maintenance planning and more informed operation. **Deliverables:** sensor-node prototype, voltage/current measurement layer, local MQTT gateway, ARKA AI dashboard, edge-AI analytics, local data storage, alert/response layer and validated pilot deployment.

5. Safety and Efficiency Analysis

ARKA AI is designed primarily as a monitoring, diagnostic and decision-support layer. Electrical measurements are collected through appropriately rated and isolated sensing interfaces selected for the monitored voltage/current range; the sensing layer is not intended to replace certified electrical protection systems. Local processing reduces dependence on continuous connectivity, while threshold validation, staged testing, redundancy and fail-safe logic help manage false alarms and sensor/communication failures. By identifying abnormal voltage/current, reduced power output and equipment degradation earlier, the system supports efficient renewable-energy generation and reduces avoidable performance losses. Any automated response would operate through suitable plant-level safety interlocks.

6. Scalability and Future Development

The modular ESP32-S3/MQTT architecture can be extended from a small pilot to multiple solar-panel and inverter zones. Future development can include higher-accuracy electrical sensing, battery/energy-storage monitoring with state-of-charge estimation, larger datasets, improved plant-specific AI models, additional environmental sensing, secure remote monitoring, fleet-level analytics and integration with existing plant SCADA systems.

7. Innovation Advantage

ARKA AI combines low-cost distributed sensing, **real-time electrical monitoring**, local edge intelligence, offline-capable monitoring, predictive pattern analysis and a unified HMI in a renewable-energy deployment. Unlike monitoring that only observes environmental conditions, ARKA AI links operating conditions to actual electrical behaviour by measuring voltage and current and presenting derived power/energy information on the dashboard. Its key advantage is adapting the intelligent sense–analyse–predict–respond architecture to solar generation while reducing dependence on continuous cloud infrastructure.

8. Visual Impact & User Journey

BEFORE: scattered/manual readings → limited visibility of voltage/current and output → delayed anomaly identification → reactive maintenance → possible performance loss.
AFTER: voltage + current + irradiance + temperature sensing → ESP32-S3 → MQTT → local gateway → ARKA AI edge analysis → live dashboard (V/A/W/Wh-kWh + trends) → anomaly/predictive alert → faster inspection & maintenance → improved visibility and optimized solar-plant performance.

Core Architecture: Solar system → Voltage/Current & Environmental Sensors → ESP32-S3 → MQTT → Local Gateway/HMI → ARKA AI Edge Analysis → Dashboard + Alerts/Response

Theme: Sustainable Energy | **Sub-theme:** Renewable Energy Harvesting & Generation Systems